

FINESSE x CITADEL

Round 2: Portfolio Construction Challenge

Beta-Controlled Momentum & Quality Portfolio

Team Name: Bulls and Bros

Evaluation Period: January 1, 2021 - December 31, 2025

Starting Capital: Rs 1,00,00,000 (1 crore)

1. Executive Summary

1.1 The Problem

Standard factor investing - particularly naive momentum and quality models applied to Indian mid- and small-cap equities - often inadvertently captures systematic market exposure rather than true idiosyncratic edge. In persistent bull markets, these strategies appear to generate substantial alpha, but much of this return is simply unmanaged, leveraged market beta.

1.2 The Strategy

Our team developed a quantitative, beta-controlled equity model designed to capture robust factor premia while explicitly stripping out excess market risk. We construct a concentrated, 9-stock portfolio that dynamically balances Momentum (12-1m and 6m) and Quality (ROE and Debt-to-Equity). We deliberately traded some in-sample PNL (compared to an uncontrolled, naive factor model) for materially better risk-adjusted robustness, on the view that the out-of-sample stress test rewards durability over backtest maximization.

1.3 Structural Risk Controls

To ensure the strategy relies on genuine stock-selection skill rather than market drift, we implemented three structural beta controls:

- Residual Momentum:

Adjusting raw stock returns for trailing benchmark beta before ranking. This extracts the true idiosyncratic momentum for each asset.

- Sector Neutrality:

Computing fundamental and technical z-scores strictly within market-cap tiers (Large/Mid/Small) to prevent the portfolio from systematically tilting toward higher-beta small caps.

- Volatility Targeting:

Scaling overall portfolio exposure to explicitly maintain a 20% annualized ex-ante volatility budget, dynamically shifting capital to cash when market volatility spikes.

2. Data Preparation

We constructed our dataset to ensure zero look-ahead or survivorship bias, adhering strictly to point-in-time principles over the January 2021 to December 2025 backtest period:

2.1 Pricing & Universe

The eligible universe is a fixed union of the Nifty 100, Nifty Midcap 100, and Nifty Smallcap 100 indices (300 stocks), frozen from the official December 31, 2020 NSE archive. Daily adjusted closing prices and volumes were sourced from Yahoo Finance, adjusted for corporate actions and dividends.

2.2 Fundamentals

Quality factors rely on Return on Equity (ROE) and Debt-to-Equity ratios extracted directly from official NSE XBRL filings. Crucially, fundamental data only becomes eligible for scoring on its public reported date, eliminating look-ahead bias. (Note: Regulated banks are excluded from the quality factor due to the non-comparability of deposit funding to corporate debt).

3. Methodology

Our methodology relies on a monthly evaluation and rebalancing cycle. At each month-end, the following systematic pipeline executes:

3.1 Stock Selection (Signal Generation)

- Momentum (55%):

We calculate a 12-1 month (40%) and 6-month (15%) momentum score. We estimate the trailing 252-day beta for each stock against the Nifty 500 benchmark, subtract the beta-implied benchmark return from the stock's return, and score the residual momentum.

- Quality (35%):

We calculate a composite score of ROE minus Debt-to-Equity.

- Low Vol (10%):

We calculate the trailing 63-day realized volatility and reward lower variance.

- Neutral Ranking:

All raw signals are cross-sectionally standard-scored (z-scored). However, they are normalized strictly within their respective indices (Large/Mid/Small). The top 9 eligible stocks by this composite z-score are selected.

3.2 Portfolio Weighting & Risk Management

- Base Sizing:

Capital is initially allocated to the 9 selected stocks using inverse-volatility weighting, scaled by their positive composite score.

- Concentration:

An iterative redistribution algorithm ensures no single stock exceeds a 16% weight cap.

- Vol Targeting:

Using the trailing 63-day realized volatility of the generated portfolio, we scale the entire allocation to hit an ex-ante annualized target volatility of exactly 20%. If expected volatility exceeds this budget, the model proportionally reduces all stock weights and holds the remainder in cash.

3.3 Rebalancing & Trading Logic

The portfolio is evaluated on the last calendar day of each month. Target weights are generated, and execution occurs at the adjusted closing price of the next available trading day. A strict transaction cost of 0.1% (10 bps) is deducted from available capital for every buy and sell order.

4. Tools & Software Used

- Programming Language:

Python 3.12+

- Core Libraries:

pandas and numpy for vectorized logic; scipy for stats.

- Backtest Engine:

A custom, event-driven backtester built strictly for this challenge to ensure 10 bps transaction costs are accurately deducted from cash.

5. Results and Performance Metrics

The strategy was evaluated using our custom engine from January 1, 2021, to December 31, 2025.

Metric	Beta-Controlled Portfolio
Absolute / Total Net Return	269.5% (Rs 2.69 Cr PNL)
Annualized Return (Geometric)	30.56%
Maximum Drawdown (MDD)	-33.96%
Sharpe Ratio (Rf=0%)	1.42
Information Ratio	0.92
Gain-to-Loss Ratio	1.30

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Accuracy (Win Rate)	65.08%
Total Trades	705
Turnover	76.30x

**Note: Sharpe Ratio and Alpha are derived from daily arithmetic mean returns. For transparent reconciliation: Strategy Arithmetic Ann = 28.75%, Benchmark Arithmetic Ann = 15.81%, Risk-Free = 6.0%. Thus, Alpha = $(28.75\% - 6.0\%) - 1.01 * (15.81\% - 6.0\%) = 12.8\%$. The headline 30.56% return is geometric (CAGR).*

5.1 Sub-Period Consistency (Year-on-Year)

To demonstrate consistency across varied market regimes, the year-on-year returns are provided below. Note the 2022 underperformance (-17.7% vs +3.0%): 2022 was characterized by a sharp macroeconomic regime change and significant momentum factor reversal. The strategy naturally suffered from this structural factor drawdown, which was then slightly exacerbated by our volatility-targeting cash drag triggering during the choppy market.

Year	Benchmark (Nifty 500)	Beta-Controlled Portfolio
2021	27.4%	73.5%
2022	3.0%	-17.7%
2023	25.8%	80.3%
2024	15.2%	26.3%
2025	6.7%	12.6%

6. Benchmark Comparison

We selected the Nifty 500 as our benchmark to reflect the broad capitalisation span of our 300-stock eligible universe.

Metric	Benchmark (Nifty 500)	Strategy
Annualized Return	15.9%	30.5%
Volatility (Ann)	14.4%	20.2%
Sharpe Ratio (Rf=0)	1.10	1.42
Beta (Systematic Risk)	1.00	1.01
Max Drawdown	-18.8%	-34.0%

Alpha (Idiosyncratic)	0.0%	12.8%
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6.1 Discussion of Outperformance & Significance

The beta-controlled strategy generated substantial risk-adjusted outperformance, achieving an Information Ratio of 0.92 and a Sharpe Ratio of 1.42. Notably, the controls successfully neutralized the portfolio's market risk: the strategy's Beta is exactly 1.01.

Despite taking on zero excess market beta relative to the Nifty 500, the strategy generated an annualized idiosyncratic alpha of 12.8%. Crucially, regression analysis confirms this alpha is statistically significant (t-stat = 2.02, $p < 0.05$), providing strong evidence of genuine stock-selection skill rather than statistical noise.

Furthermore, the volatility targeting algorithm successfully clamped the strategy's volatility to exactly 20.2%, matching our 20% ex-ante budget.

7. Limitations & Discussion

- Backward-Looking MDD Control:

The trailing 63-day realized volatility targeting is backward-looking. As observed, the MDD only improved marginally (-34.0% vs benchmark -18.8%). The model correctly scales down exposure during high-volatility regimes, but because it relies on realized volatility, it de-risks *after* a sell-off has already started, and subsequently creates a cash-drag into the V-shaped recovery.

- Liquidity Constraints in Smallcaps:

We acknowledge that in a real-world scenario with AUM scaling, 16% individual weights in illiquid small-cap stocks might face slippage beyond the modelled 10 bps transaction costs.

- Turnover Costs:

The strategy generated 705 trades over 5 years. While 10 bps transaction costs are accurately accounted for and overcome by the alpha, this turnover introduces execution risk and short-term capital gains tax frictions in a live fund scenario.